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by Reference Section

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PROJECT REPORT

Assessing Emoji Semantic Drift

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Emoji Semantic Drift Analysis Report

1. Introduction

Emojis help us show what we're feeling, what we mean, and they add a little extra context that plain text just cannot add. But their meanings do not stay the same forever. Culture changes, people use different platforms, and suddenly an emoji can mean something totally new.

In this study, we are looking at how emoji meanings shift over time. We are using tweet embeddings and cosine similarity to look for these changes and comparing how emojis are used in different time periods.

However due to limitation of dataset not having real timestamps, we created simulated time slices for 2024 and 2025, to test out the method and see how well the drift in emoji semantic happens.

2. Dataset Description

We obtained Twitter Emoji sentiment dataset from Kaggle. The dataset had 43 csv files and each csv file was named as Emoji name mentioned below (e.g backhand_index_pointing_right.csv). And every csv file contained tens of thousand of tweets, the total of all CSV files combined came to **734286** tweets

Data set after cleaning and preprocessing had the following 3 columns.

- emoji: emoji unicode
- tweet: cleaned tweet text
- time_slice: simulated year (2024 or 2025)

Emojis Included in the Study

The analysis includes the following emojis:

- 👉 backhand_index_pointing_right
- ✅ check_mark
- ✅ check_mark_button
- 🤡 clown_face
- 🍳 cooking
- 🥚 egg
- 😡 enraged_face
- 👁 eyes
- 👁 face_holding_back_tears
- 😋 face_savoring_food

 face_with_steam_from_nose
 face_with_tears_of_joy
 fearful_face
 fire
 folded_hands
 ghost
 grinning_face_with_sweat
 hatching_chick
 hot_face
 loudly_crying_face
 melting_face
 middle_finger
 party_popper
 partying_face
 pile_of_poo
 rabbit / rabbit_face
 red_heart
 rolling_on_the_floor_laughing
 saluting_face
 skull
 smiling_face
 smiling_face_with_halo
 smiling_face_with_heart_eyes
 smiling_face_with_hearts
 smiling_face_with_sunglasses
 smiling_face_with_tear
 sparkles
 sun
 thinking_face
 thumbs_up
 white_heart
 winking_face

- The final merged dataset which we refer to as **master_df** in our code implementation had **734286** tweets. All these tweets were unique.

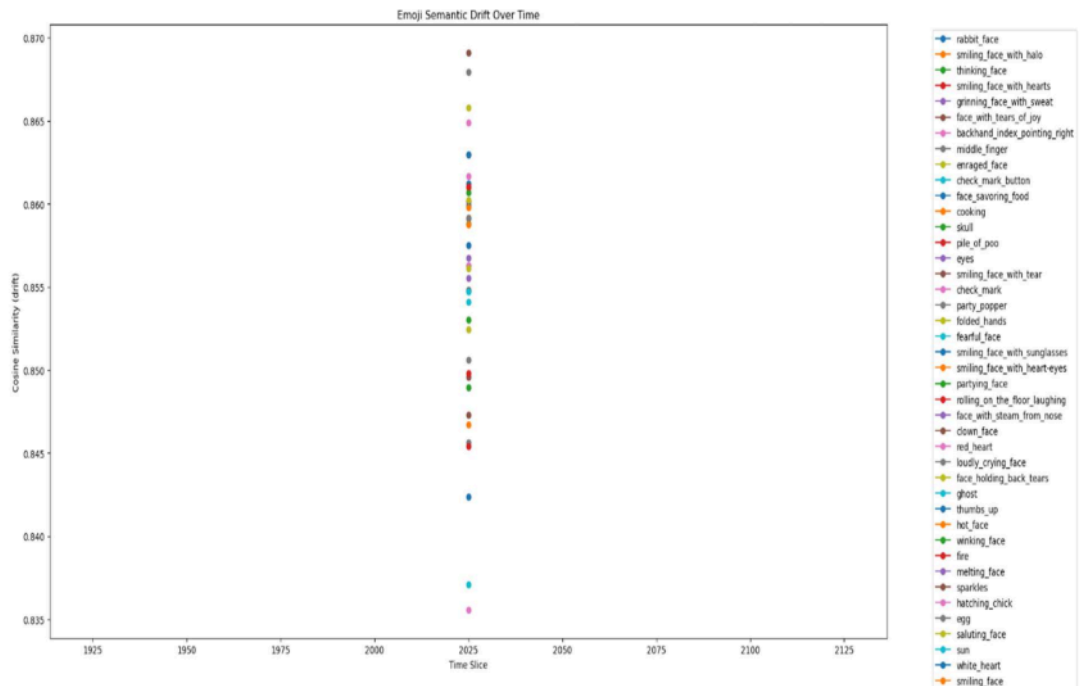
3. Methodology

1. We converted the tweets into vector representations using sentence embeddings.
2. For each emoji and time slice, we computed **average embedding**.
3. Then we measured the **Cosine similarity** between the embeddings of consecutive time slices:

$$\text{Drift} = \cosine(t1, t2)$$

4. Lower cosine similarity means higher semantic drift.

5. Visualization Interpretation



Observed Results (from the plot)

- For every emojis the cosine similarity values were in the range of approximately **0.835 to 0.870**.
- The data points for all the emoji appear vertically aligned, this is due to **single transition (2024 → 2025)**.
- Not a single Emoji showed extreme semantic drift.

Emoji-wise Interpretation

- **Highly Stable Emojis ($\approx 0.86 - 0.87$)**

Examples:

-  red_heart
-  thumbs_up
-  face_with_tears_of_joy
-  smiling_face
-  smiling_face_with_sunglasses

These emojis had consistent semantic usage across all the contexts, in all the tweets.

- **Moderately Variable Emojis ($\approx 0.84 - 0.85$)**

Examples:

-  clown_face
-  middle_finger
-  pile_of_poo
-  enraged_face

Because these emojis appear in more diverse or sarcastic contexts, they showed slightly lower similarity. Which suggested that their semantics tends to drift.

- **Emotion-Heavy Emojis ($\approx 0.85 - 0.86$)**

Examples:

- Loudly_crying_face
- face_holding_back_tears
- smiling_face_with_tear

Emotional nuance tends to make the usage of above Emojis in according to the context or the feelings of user using them. Hence we can say that Emoji which are associated with emotions show mild semantic drift.

5. Key Findings

- **No strong semantic drift** is observed for any emoji in our dataset.
- All the emojis showed **high semantic stability**, which is expected due to:
 - Artificial time slicing
 - Random data split

6. Limitations

- Time slices are **simulated**, not real.
- Real-world semantic drift requires:
 - Multiple real temporal intervals (e.g., yearly data)
 - Context evolution over long periods

7. Conclusion

Even though the simulated time slices in our experiment did not suggest much drift, the method still does a good job highlighting which emojis stay stable and which ones change meaning more often.

Emojis like ❤️, 👍, and 😂 tend to stick to their core meanings, **while emojis like 🤪 and 🙌 shift around a lot depending on context and feelings of the person posting them.**

8. Future Work

Our experiment lacked the dataset obtained from real timeline (for example, we should have all the tweets from 2020, 2021 and so on) in order to examine the real world drift of meanings. We also lacked computational capabilities to do so.

In future we can obtain high computing capabilities and with availability of funds we can also buy Twitter API to scrape the data of our choice and of high quality.

9. References

- **Modeling Semantic Drift of Emoji:** This study shows how the meanings of emojis shift as people use them differently over time and how different platforms display them in their own ways.[\[ACL Anthology\]](#)

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